

CSA0609-DESIGN AND ANALYSIS OF ALGORITHMS

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Exp-06: Champagne Tower

Aim:

To determine the amount of champagne present in a specified glass of a champagne tower after pouring a given number of cups.

Algorithm:

1. Read the number of poured cups.
2. Read the required row and glass position.
3. Initialize the champagne tower.
4. Pour the champagne into the top glass.
5. If a glass contains more than one cup, distribute the excess equally between the two glasses below it.
6. Continue the process for all rows.
7. Find the amount in the required glass.
8. Display the result and complexity.

Source Code:

```
poured = int(input("Enter number of cups poured: "))
```

```
query_row = int(input("Enter row number: "))
```

```
query_glass = int(input("Enter glass number: "))
```

```
tower = [[0.0 for _ in range(query_row + 2)]
```

```
         for _ in range(query_row + 2)]
```

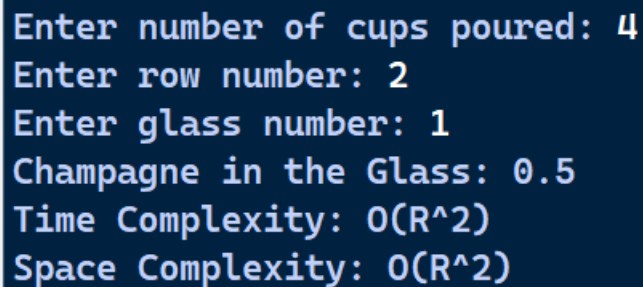
```
tower[0][0] = poured
```

```
for i in range(query_row):
    for j in range(i + 1):
        if tower[i][j] > 1:
            excess = (tower[i][j] - 1) / 2
            tower[i + 1][j] += excess
            tower[i + 1][j + 1] += excess

answer = min(1, tower[query_row][query_glass])

print("Champagne in the Glass:", answer)
print("Time Complexity: O(R^2)")
print("Space Complexity: O(R^2)")
```

Output:

A screenshot of a terminal window with a dark blue background and light blue text. It shows the input and output of the program. The inputs are: 'Enter number of cups poured: 4', 'Enter row number: 2', and 'Enter glass number: 1'. The outputs are: 'Champagne in the Glass: 0.5', 'Time Complexity: O(R^2)', and 'Space Complexity: O(R^2)'.

```
Enter number of cups poured: 4
Enter row number: 2
Enter glass number: 1
Champagne in the Glass: 0.5
Time Complexity: O(R^2)
Space Complexity: O(R^2)
```

Result:

The amount of champagne present in the specified glass was successfully calculated.